

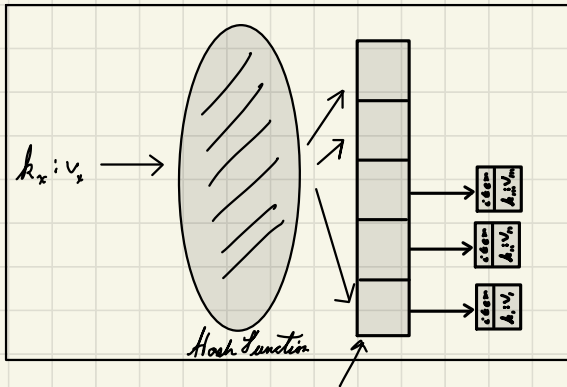
Maps



Maps

- Python dictionaries
- Collection of key-value pairs
- Rules:
 - Each key must be unique.
 - Each key maps to exactly one value. (Not necessarily unique)
- Dynamic in size (grows as you add k-v pairs; shrinks as you remove)
- Really efficient operations: deletion, insertion, lookup $\rightarrow O(1)$ (due to hash table implementation)

Operation	Definition/Description	Example code
<code>m = Map()</code>	Creates an empty map	<code>m = {}</code>
<code>m[k] = v</code>	Adding / updating	<code>m['a'] = 10</code>
<code>m[k]</code>	Lookup / retrieval (k is mapped to v)	<code>value = m['a']</code>
<code>del m[k]</code>	Removes k-v pair. Also raises key error if key	<code>del m['a']</code>
<code>len(m)</code>	Returns # of k-v pairs	<code>✓</code>
<code>iter(m)</code>	Returns iterator of the map keys	<code>for key in m</code>



$O(1)$

$\text{len}(k_x) * 3 = 9$
`"seb"` $\rightarrow$ $\begin{bmatrix} H \\ f \end{bmatrix} \rightarrow 437$
`"set"` $\rightarrow \begin{bmatrix} H \\ f \end{bmatrix} \rightarrow 25$